

CS & IT ENGINEERING

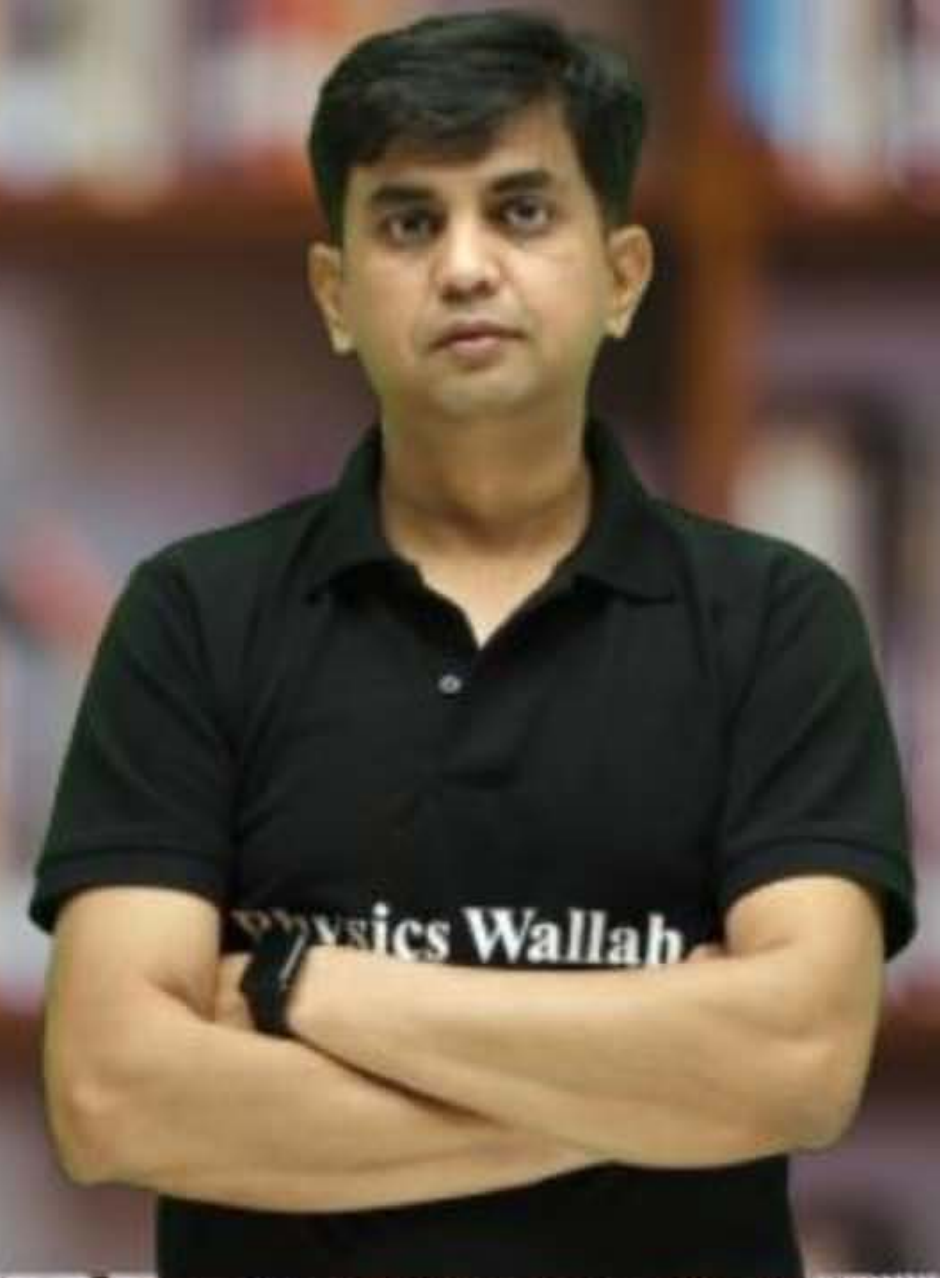


Computer Network

IPv4 Address

Lecture No. - 07

By - Abhishek Sir





Recap of Previous Lecture



Topic

Subnetting (Fixed length)





Topics to be Covered



Topic

VLSM



ABOUT ME



Hello, I'm **Abhishek**

- GATE CS AIR - 96
- M.Tech (CS) - IIT Kharagpur
- 12 years of GATE CS teaching experience

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Classless network (Network Address : 90 . 152 . 0 . 0 / 14), with 5-bit subnetting.

Network Address : 90 . 152 . 0 . 0 / 14

Broadcast Address : 90 . 155 . 255 . 255 / 14
(Network Directed)

100 | 1000

100 | 1011

Before Subnetting :-

First Host IP Address : 90 . 152 . 0 . 1

Last Host IP Address : 90 . 155 . 255 . 254

Network Size = $(2^{18} - 2)$ hosts

After 5-bit Subnetting :-

14 bit NetID 5 bit SubnetID 13 bit HostID

Sub-network Address : 90.100110 _ _ . _ _ _ 00000.00000000 / 19

First Host IP Address : 90.100110 _ _ . _ _ _ 00000.00000001 / 19

Last Host IP Address : 90.100110 _ _ . _ _ _ 11111.11111110 / 19

Broadcast Address : 90.100110 _ _ . _ _ _ 11111.11111111 / 19
(Sub-network Directed)

1st Sub-network Address : 90.10011000.00000000.0/19

: 90.152.0.0/19

First Host IP Address : 90.10011000.00000001.1/19

: 90.152.0.~~0~~/19
|

Last Host IP Address : 90.10011000.00011111.254/19

: 90.152.31.~~0~~/19
254

1st Subnet Broadcast Address : 90.10011000.00011111.255/19

(Sub-network Directed) : 90.152.31.~~0~~/19
255

2nd Sub-network Address : 90.10011000.00100000.0/19

: 90.152.32.0/19

First Host IP Address : 90.10011000.00100000.1/19

: 90.152.32.~~0~~/19

Last Host IP Address : 90.10011000.0011111.254/19

: 90.152.63.~~0~~/19

2nd Subnet Broadcast Address : 90.10011000.0011111.255/19

(Sub-network Directed) : 90.152.63.~~0~~/19

254
255

Last Sub-network Address : 90.100110 . 00000.0 / 19

: 90.155.224.0 / 19

First Host IP Address : 90.100110 . 00000.1 / 19

: 90.155.224.~~0~~ / 19

Last Host IP Address : 90.100110 . 11111.254 / 19

: 90.155.255.~~0~~ / 19

254

Last Subnet Broadcast Address : 90.100110 . 11111.255 / 19

(Sub-network Directed) : 90.155.255.~~0~~ / 19

255

First Sub-network Address : 90.152.0.0 / 19
 Second Sub-network Address : 90.152.32.0 / 19
 Third Sub-network Address : 90.152.64.0 / 19
 Fourth Sub-network Address : 90.152.96.0 / 19
 152.128
 152.160
 152.192
 152.224
 153.0
 ⋮
 31th Sub-network Address : 90.155.192.0 / 19
 Last (32nd) Sub-network Address : 90.155.224.0 / 19



Topic : Subnetting

Suppose,

Net ID field size is 27 bits

Host ID field size is 5 bits

→ 2 bit subnetting

Subnet Mask



discontinuous one's

Host ID

0 0 0 0 0

0 0 0 0 1

0 0 0 1 0

0 0 0 1 1

0 0 1 0 0

0 0 1 0 1

0 0 1 1 0

0 0 1 1 1

0 1 0 0 0

0 1 0 0 1

0 1 0 1 0

0 1 0 1 1

0 1 1 0 0

0 1 1 0 1

0 1 1 1 0

0 1 1 1 1

1 0 0 0 0

1 0 0 0 1

1 0 0 1 0

1 0 0 1 1

1 0 1 0 0

1 0 1 0 1

1 0 1 1 0

1 0 1 1 1

1 1 0 0 0

1 1 0 0 1

1 1 0 1 0

1 1 0 1 1

1 1 1 0 0

1 1 1 0 1

1 1 1 1 0

1 1 1 1 1



Topic : VLSM



- VLSM : Variable Length Subnet Mask
- Variable Length Network prefixes
[Unlike fixed length prefixing]
- Allow variable size sub-networks (subnets)
[efficient utilization of IP address space]



Topic : VLSM

Suppose,

Net ID field size is 27 bits

Host ID field size is 5 bits

Host ID (5 bits)

0	0	0	0	0
0	0	0	0	1
0	0	0	1	0
0	0	0	1	1
0	0	1	0	0
0	0	1	0	1
0	0	1	1	0
0	0	1	1	1
0	1	0	0	0
0	1	0	0	1
0	1	0	1	0
0	1	0	1	1
0	1	1	0	0
0	1	1	0	1
0	1	1	1	0
0	1	1	1	1

1	0	0	0	0
1	0	0	0	1
1	0	0	1	0
1	0	0	1	1
1	0	1	0	0
1	0	1	0	1
1	0	1	1	0
1	0	1	1	1
1	1	0	0	0
1	1	0	0	1
1	1	0	1	0
1	1	0	1	1
1	1	1	0	0
1	1	1	0	1
1	1	1	1	0
1	1	1	1	1





Topic : VLSM

Suppose,

Net ID field size is 27 bits

Host ID field size is 5 bits

Host ID (5 bits)

0	0	0	0	0
0	0	0	0	1
0	0	0	1	0
0	0	0	1	1
0	0	1	0	0
0	0	1	0	1
0	0	1	1	0
0	0	1	1	1
0	1	0	0	0
0	1	0	0	1
0	1	0	1	0
0	1	0	1	1
0	1	1	0	0
0	1	1	0	1
0	1	1	1	0
0	1	1	1	1

1	0	0	0	0
1	0	0	0	1
1	0	0	1	0
1	0	0	1	1
1	0	1	0	0
1	0	1	0	1
1	0	1	1	0
1	0	1	1	1
1	1	0	0	0
1	1	0	0	1
1	1	0	1	0
1	1	0	1	1
1	1	1	0	0
1	1	1	0	1
1	1	1	1	0
1	1	1	1	1





Topic : VLSM

Suppose,

Net ID field size is 27 bits

Host ID field size is 5 bits

Host ID (5 bits)

0	0	0	0	0
0	0	0	0	1
0	0	0	1	0
0	0	0	1	1
0	0	1	0	0
0	0	1	0	1
0	0	1	1	0
0	0	1	1	1
0	1	0	0	0
0	1	0	0	1
0	1	0	1	0
0	1	0	1	1
0	1	1	0	0
0	1	1	0	1
0	1	1	1	0
0	1	1	1	1

1	0	0	0	0
1	0	0	0	1
1	0	0	1	0
1	0	0	1	1
1	0	1	0	0
1	0	1	0	1
1	0	1	1	0
1	0	1	1	1
1	1	0	0	0
1	1	0	0	1
1	1	0	1	0
1	1	0	1	1
1	1	1	0	0
1	1	1	0	1
1	1	1	1	0
1	1	1	1	1



Classless network (Network Address : $120.150.0.0 / 16$)

[Wishes to divide the network into 3 subnets, one with 50% hosts and remaining two with 25% hosts.]

Network Address : $120.150.0.0 / 16$

Broadcast Address : $120.150.255.255 / 16$
(Network Directed)

Before Subnetting :-

First Host IP Address : $120.150.0.1 / 16$

Last Host IP Address : $120.150.255.254 / 16$

Network Size = $(2^{16} - 2)$ hosts

Classless network (Network Address : 120 . 150 . 0 . 0 / 16)

120 . 150 . _____ / 16
 16 bit Host ID

Subnet-1 Address : 120 . 150 . 0 0 0 0 0 0 0 0 . 0 0 0 0 0 0 0 0 / 17
 [for 50% hosts]
 15 bit Host ID

Subnet-2 Address : 120 . 150 . 1 0 0 0 0 0 0 0 . 0 0 0 0 0 0 0 0 / 18
 [for 25% hosts]
 14 bit Host ID

Subnet-3 Address : 120 . 150 . 1 1 0 0 0 0 0 0 . 0 0 0 0 0 0 0 0 / 18
 [for 25% hosts]
 14 bit Host ID

Classless network (Network Address : 120 . 150 . 0 . 0 / 16)

120 . 150 . _____ / 16
 16 bit Host ID

Subnet-1 Address : 120 . 150 . 1 00000000 . 000000000000 / 17
 [for 50% hosts]
 15 bit Host ID

Subnet-2 Address : 120 . 150 . 0 0 00000000 . 000000000000 / 18
 [for 25% hosts]
 14 bit Host ID

Subnet-3 Address : 120 . 150 . 0 1 00000000 . 000000000000 / 18
 [for 25% hosts]
 14 bit Host ID

Subnet-1 Address : 120 . 150 . 0 . 0 / 17

Subnet-2 Address : 120 . 150 . 128 . 0 / 18

Subnet-3 Address : 120 . 150 . 192 . 0 / 18

Subnet-1 Address : 120 . 150 . 128 . 0 / 17

Subnet-2 Address : 120 . 150 . 0 . 0 / 18

Subnet-3 Address : 120 . 150 . 64 . 0 / 18

Subnet-1 Address : 120 . 150 . 0 . 0 / 17

Subnet-2 Address : 120 . 150 . 192 . 0 / 18

Subnet-3 Address : 120 . 150 . 128 . 0 / 18

Subnet-1 Address : 120 . 150 . 128 . 0 / 17

Subnet-2 Address : 120 . 150 . 64 . 0 / 18

Subnet-3 Address : 120 . 150 . 0 . 0 / 18

#Q. A company has a class C network address of “204 . 204 . 204 . 0” . It wishes to have three subnets, one with 100 hosts and two with 50 hosts each. Which one of the following options represents a feasible set of subnet address/subnet mask pairs?

- (A) 204 . 204 . 204 . 128 / 255 . 255 . 255 . 192
 204 . 204 . 204 . 0 / 255 . 255 . 255 . 128
 204 . 204 . 204 . 64 / 255 . 255 . 255 . 128
- (B) 204 . 204 . 204 . 0 / 255 . 255 . 255 . 192
 204 . 204 . 204 . 192 / 255 . 255 . 255 . 128
 204 . 204 . 204 . 64 / 255 . 255 . 255 . 128
- (C) 204 . 204 . 204 . 128 / 255 . 255 . 255 . 128
 204 . 204 . 204 . 192 / 255 . 255 . 255 . 192
 204 . 204 . 204 . 224 / 255 . 255 . 255 . 192
- (D) 204 . 204 . 204 . 128 / 255 . 255 . 255 . 128
 204 . 204 . 204 . 64 / 255 . 255 . 255 . 192
 204 . 204 . 204 . 0 / 255 . 255 . 255 . 192

[GATE-IT-2005, 2 Marks]

IIT-B, H.W.



Class C network (Network Address : 204 . 204 . 204 . 0)

204 . 204 . 204 . _ _ _ _ _

Subnet-1 Address : 204 . 204 . 204 . _ _ _ _ _
[for 100 host]

Subnet Mask = 255 . 255 . 255 .

Subnet-2 Address : 204 . 204 . 204 . _ _ _ _ _
[for 50 host]

Subnet Mask = 255 . 255 . 255 .

Subnet-3 Address : 204 . 204 . 204 . _ _ _ _ _
[for 50 host]

Subnet Mask = 255 . 255 . 255 .



Subnet-1 Address : 204 . 204 . 204 .

Subnet Mask = 255 . 255 . 255 .

Subnet-2 Address : 204 . 204 . 204 .

Subnet Mask = 255 . 255 . 255 .

Subnet-3 Address : 204 . 204 . 204 .

Subnet Mask = 255 . 255 . 255 .

Subnet-1 Address : 204 . 204 . 204 .

Subnet Mask = 255 . 255 . 255 .

Subnet-2 Address : 204 . 204 . 204 .

Subnet Mask = 255 . 255 . 255 .

Subnet-3 Address : 204 . 204 . 204 .

Subnet Mask = 255 . 255 . 255 .



2 mins Summary



Topic

VLSM



THANK - YOU

